

№3

25 2025

‘ ‘

LaTeX,

.

1. LaTeX
- 2.
- 3.
- 4.
- 5.

$$c \quad \backslash$$

$$: - \quad - \quad : \$y = mx + c\$ - \quad - \quad : \backslash[y = mx +$$

```

\documentclass{article}
\usepackage[T1]{fontenc}
\usepackage{amsmath}
\usepackage{amsfonts}
\usepackage{bm}
\begin{document}

\section{Mathematics Typing - Main Learning}

\subsection{1. Basic Math Modes}
Inline math:  $y = mx + c$  \\
Display math:

$$y = mx + c$$


\subsection{2. Superscripts and Subscripts}
 $x^2$ ,  $a_n$ ,  $x_i^2$ ,  $n_{k-1}^{m+1}$ 

\subsection{3. Greek Letters and Functions}
 $\alpha$ ,  $\beta$ ,  $\theta$ ,  $\Gamma$ ,  $\sin\theta$ ,  $\log x$ 

\subsection{4. Integration Example}

$$\int_{-\infty}^{\infty} e^{-x^2} \, dx$$

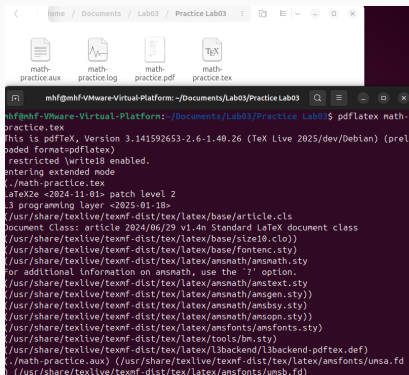

\subsection{5. Aligned Equations}
\begin{align*}
a &= b + c \\
x &= y - z
\end{align*}

```

math-practice.tex

:

-
-
-
-



The screenshot shows a terminal window titled "mhf@mhf-VMware-Virtual-Platform: ~/Documents/Lab03/Practice Lab03". The terminal displays the output of the command `pdflatex math-practice.tex`. The output indicates that pdflatex, Version 3.141592653-2.6-1.40.26 (TeX Live 2025/dev/Debian) is being used. It shows the document class, font settings, and the inclusion of various LaTeX packages like `amsmath`, `amstext`, `ansnath`, `ansgen`, `ansbsy`, `ansopn`, and `ansfont`. The terminal also shows the path to the `math-practice.aux` file and the `math-practice.pdf` output file.

```
mhf@mhf-VMware-Virtual-Platform: ~/Documents/Lab03/Practice Lab03$ pdflatex math-practice.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.26 (TeX Live 2025/dev/Debian) (prel
oaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
./math-practice.tex
LaTeX2e <2024-11-01> patch level 2
 3 programming layer <2025-01-18>
(/usr/share/texlive/texmf-dist/tex/latex/base/article.cls
Document Class: article 2024/06/29 v1.4n Standard LaTeX document class
(/usr/share/texlive/texmf-dist/tex/latex/base/size10.clo))
(/usr/share/texlive/texmf-dist/tex/latex/base/fontenc.sty)
(/usr/share/texlive/texmf-dist/tex/latex/ansnath/ansmath.sty
For additional information on ansnath, use the '?' option.
(/usr/share/texlive/texmf-dist/tex/latex/ansnath/amstext.sty
(/usr/share/texlive/texmf-dist/tex/latex/ansnath/ansgen.sty))
(/usr/share/texlive/texmf-dist/tex/latex/ansnath/ansbsy.sty)
(/usr/share/texlive/texmf-dist/tex/latex/ansnath/ansopn.sty))
(/usr/share/texlive/texmf-dist/tex/latex/ansfont/ansfont.sty)
(/usr/share/texlive/texmf-dist/tex/latex/tools/bm.sty)
(/usr/share/texlive/texmf-dist/tex/latex/l3backend/l3backend-pdfTeX.def)
./math-practice.aux) (/usr/share/texlive/texmf-dist/tex/latex/ansfont/ansfont.
(/usr/share/texlive/texmf-dist/tex/latex/ansfont/ansfont.f
(/usr/share/texlive/texmf-dist/tex/latex/ansfont/ansfont.f
```

:

pdflatex math-practice.tex

-

```
%
\int_{-\infty}^{\infty} e^{-x^2} \, dx
\]

\subsection{5. Aligned Equations}
\begin{align*}
a &= b + c \\
x &= y - z \\
Q_{n,k} &= Q_{n-1,k} + Q_{n-1,k-1}
\end{align*}

\subsection{6. Numbered Equation and Matrices}
\begin{equation}
e^{i\pi} + 1 = 0
\end{equation}

\[
\begin{pmatrix}
1 & 2 \\
3 & 4
\end{pmatrix}
\end{pmatrix}
\]
```

exercises.tex 4 :

```
\documentclass{article}
\usepackage[T1]{fontenc}
\usepackage{amsmath}
\usepackage{amsfonts}
\begin{document}

\section{Exercises from Section 3}

\subsection{Exercise 1: Inline vs Display Math}
Take the same formula and try it both ways: \\
Inline:  $f(x) = x^2 + y^2$  \\
Display:
\[
f(x) = x^2 + y^2
\]

\subsection{Exercise 2: Greek Letter Practice}
Try these Greek Letters: \\
 $\Delta$ ,  $\Delta$ ,  $\sigma$ ,  $\Sigma$ ,  $\phi$ ,  $\Phi$ 
```

:

pdflatex exercises.tex

```
mhf@mhf-VMware-Virtual-Platform:~/Documents/Lab03/Exercises Lab03$ pdflatex exercises.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.26 (TeX Live 2025/dev/Debian) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./exercises.tex
LaTeX2e <2024-11-01> patch level 2
L3 programming layer <2025-01-18>
(/usr/share/texlive/texmf-dist/tex/latex/base/article.cls
Document Class: article 2024/06/29 v1.4n Standard LaTeX document class
(/usr/share/texlive/texmf-dist/tex/latex/base/size10.clo))
(/usr/share/texlive/texmf-dist/tex/latex/base/fontenc.sty)
(/usr/share/texlive/texmf-dist/tex/latex/amsmath/amsmath.sty
For additional information on amsmath, use the '?' option.
(/usr/share/texlive/texmf-dist/tex/latex/amsmath/amstext.sty
(/usr/share/texlive/texmf-dist/tex/latex/amsmath/amsgen.sty))
(/usr/share/texlive/texmf-dist/tex/latex/amsmath/amsbsy.sty)
(/usr/share/texlive/texmf-dist/tex/latex/amsmath/amsopn.sty))
(/usr/share/texlive/texmf-dist/tex/latex/amsfonts/amsfonts.sty)
(/usr/share/texlive/texmf-dist/tex/latex/l3backend/l3backend-pdfTeX.def)
(./exercises.aux) (/usr/share/texlive/texmf-dist/tex/latex/amsfonts/umsa.fd)
(/usr/share/texlive/texmf-dist/tex/latex/amsfonts/umlsb.fd)
Underfull \hbox (badness 10000) in paragraph at lines 22--25
```

. 5:

:

/ : x^2 , a_n

: α , β , Γ

: $\sin \theta$, $\log x$

: $\int e^{-x^2} dx$

: pmatrix, bmatrix

: $\mathbf{B}$, $\mathbb{R}$, $\mathbb{C}$

1 Exercises from Section 3

1.1 Exercise 1: Inline vs Display Math

Take the same formula and try it both ways:

Inline: $f(x) = x^2 + y^2$

Display:

:

$$\frac{a}{b} -$$

$$\Sigma, \Pi -$$

$$\lim, \int -$$

$$\backslash\mathrm{begin}\{\mathrm{align}^*\} -$$

$$\backslash\mathrm{begin}\{\mathrm{pmatrix}\} -$$

$$\mathbf{A} -$$

:

4

1 Mathematics Typing - Main Learning

1.1 1. Basic Math Modes

Inline math: $y = mx + c$

Display math:

$$y = mx + c$$

1.2 2. Superscripts and Subscripts

x^2 , a_n , x_i^2 , n_{k-1}^{m+1}

1.3 3. Greek Letters and Functions

α , β , θ , Γ , $\sin \theta$, $\log x$

№3:

LaTeX

amsmath

LaTeX

.